

FLASH EVENT - HIPO

Fresh Concrete Blowout from Pipeline

Description

During the pouring of concrete for a slab, fresh concrete blew out from displacement joints along the pipelines throughout the entire concreting process area.

The brackets used to tie down the pipes were found to be distorted, and some anchored bolts came off from the brackets due to the drastic force.

Consequences

Real consequences

No injury as the exclusion area was respected

Potential consequences

Severe injuries in case of people hit by pipping or by concrete projection

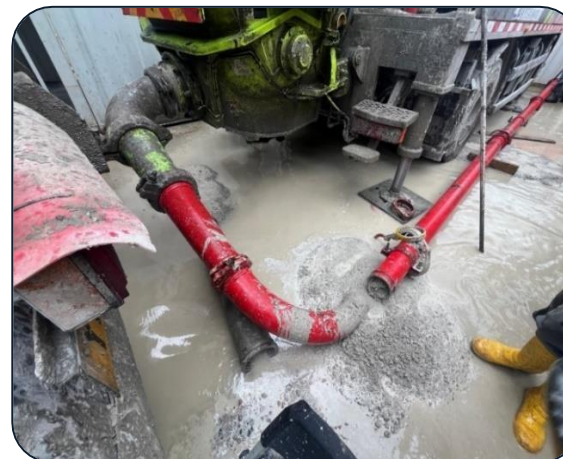
Causes of the event

Direct cause

- Mixed use of concrete pipe connectors, which are incompatible (low and high pressure bolts)

Root causes

- Insufficient planning for the layout of concrete pipeline arrangements, resulting in numerous bends along the lengthy pipeline
- Operator's insufficient skills and experience in adjusting the operating pressure of the concrete pump
- Concrete mix design issue (sticky and low fluidity)



Concreting process area



Damaged anchored bolts



Prevention measures

- Prioritize the direct-feed concreting method, and If not feasible, minimize the length of concrete pipeline and number of bends
- Ensure the use of proper types of concrete pipe connectors according to concrete plan
- When relevant, inform concrete supplier with detailed concreting arrangements (bends, lengths), and maintain effective coordination during concreting process
- Ensure that all necessary competencies for achieving the task are mobilised

CAUSES & LINES OF DEFENSE



Managerial Behaviour

- Decision not to arrange direct feed that would have disrupted other activities

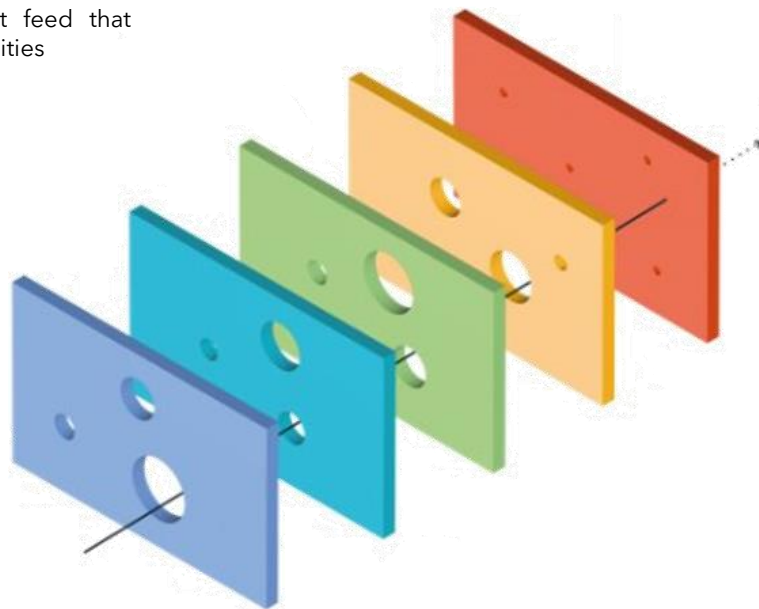


Life Saving Rules

- ✓ Exclusion zone near the concrete pipeline as been respected by personnel

Technical

- Incompatible concrete lines connectors
- Lot of bends in the concrete lines
- Concrete mix sticky and low fluidity



Human

Organisation

- Operator's insufficient skills and experience in adjusting the operating pressure of the concrete pump.